Dividing by Two-Digit Numbers

Find easy parts of the divisor. Take them out. Add.

Easy parts of 24

$24 \times 10 = 240$

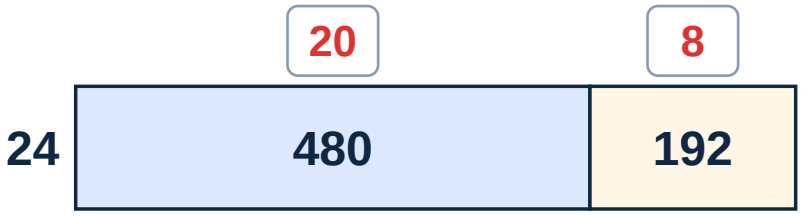
$24 \times 5 = 120$

$24 \times 20 = 480$

$24 \times 2 = 48$

Times 10 is easy.
Double it for 20.
Half of 10 is 5.

672 ÷ 24



$480 \div 24 = 20$

$192 \div 24 = 8$

$20 + 8 = 28$

Take out 20 groups of 24 (480). 192 is left. 192 is 8 groups of 24.

Not even? Write what is left

$500 \div 24 = 20$ R 20

20 groups of 24 is 480. 20 are left, and 20 is less than 24.

Your turn!

$384 \div 16 =$ 24

For teachers

Goal: divide by two-digit numbers with the area model and partial quotients (5.NBT.6), like packet 59 with a bigger side. Say "What easy part of 24 fits? 10 groups? 20 groups?" Any easy parts work; fewer parts is faster.

Dividing by Two-Digit Numbers: Practice

Easy parts first: 10 groups, 20 groups, 5 groups.

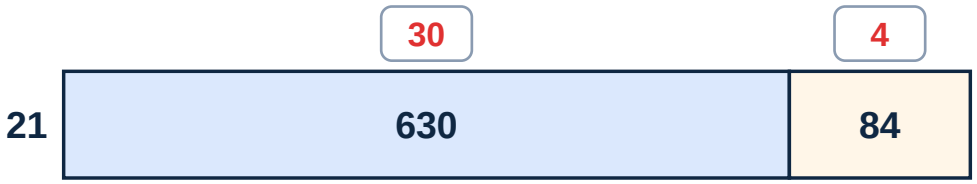
Part 1 Easy parts Times 10, times 20, times 5.

1)	32 × 10 = 320	2)	17 × 20 = 340	3)	47 × 5 = 235
4)	23 × 30 = 690	5)	41 × 10 = 410	6)	28 × 5 = 140
7)	19 × 20 = 380	8)	36 × 10 = 360	9)	14 × 30 = 420
10)	52 × 5 = 260	11)	27 × 20 = 540	12)	38 × 10 = 380

Dividing by Two-Digit Numbers: Practice (continued)

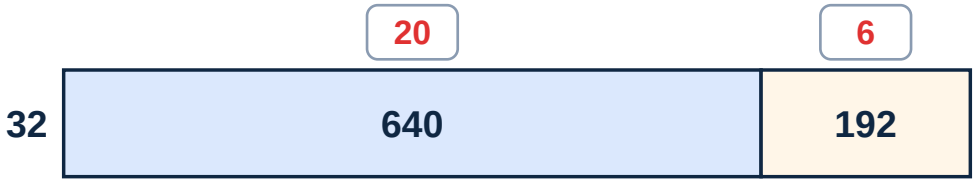
Part 2 The area model The parts are inside. Find each side on top.

13) **714 ÷ 21 = 34**



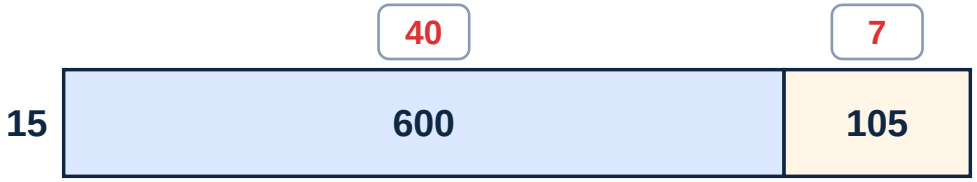
30 + 4 = 34

14) **832 ÷ 32 = 26**



20 + 6 = 26

15) **705 ÷ 15 = 47**

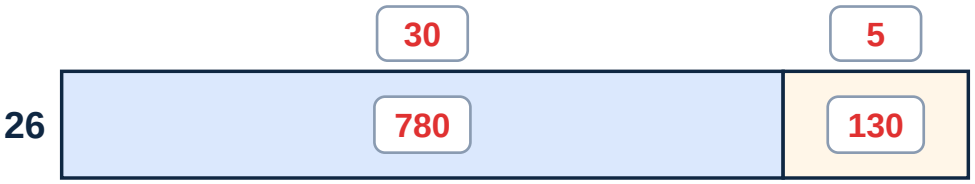


40 + 7 = 47

Dividing by Two-Digit Numbers: Practice (continued)

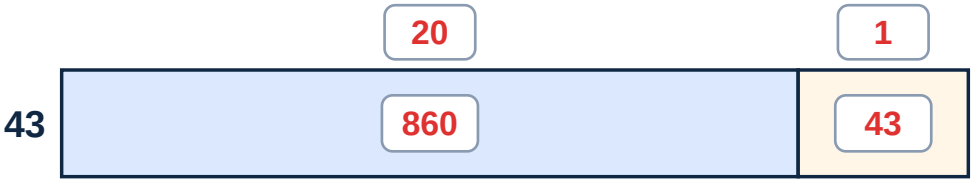
Part 3 Pick your own parts Write the parts inside, then each side on top.

16) **910 ÷ 26 = 35**



30 + 5 = 35

17) **903 ÷ 43 = 21**



20 + 1 = 21

18) **972 ÷ 18 = 54**



50 + 4 = 54

Dividing by Two-Digit Numbers: Practice (continued)

Part 4 Divide each part Then add.

19) 925 = 750 + 175

750 ÷ 25 = 30

175 ÷ 25 = 7

30 + 7 = 37

20) 744 = 620 + 124

620 ÷ 31 = 20

124 ÷ 31 = 4

20 + 4 = 24

21) 928 = 800 + 128

800 ÷ 16 = 50

128 ÷ 16 = 8

50 + 8 = 58

22) 966 = 840 + 126

840 ÷ 42 = 20

126 ÷ 42 = 3

20 + 3 = 23

23) 988 = 910 + 78

910 ÷ 13 = 70

78 ÷ 13 = 6

70 + 6 = 76

24) 986 = 680 + 306

680 ÷ 34 = 20

306 ÷ 34 = 9

20 + 9 = 29

Part 5 Divide No help this time.

25) 672 ÷ 21 = 32

26) 644 ÷ 14 = 46

27) 594 ÷ 33 = 18

28) 624 ÷ 26 = 24

29) 663 ÷ 17 = 39

30) 720 ÷ 45 = 16

Dividing by Two-Digit Numbers: Practice (continued)

Part 5 Divide (continued)

31)	876 ÷ 12 = 73	32)	798 ÷ 38 = 21	33)	621 ÷ 23 = 27
34)	714 ÷ 51 = 14	35)	836 ÷ 19 = 44	36)	792 ÷ 36 = 22
37)	868 ÷ 28 = 31	38)	870 ÷ 15 = 58	39)	836 ÷ 44 = 19

Part 6 Bigger numbers No help this time.

40)	5,544 ÷ 24 = 231	41)	5,356 ÷ 13 = 412

Dividing by Two-Digit Numbers: Practice (continued)

Part 6 Bigger numbers (continued)

42) 4,992 ÷ 32 = 156

43) 7,245 ÷ 21 = 345

44) 6,016 ÷ 47 = 128

45) 6,000 ÷ 16 = 375

46) 7,245 ÷ 35 = 207

47) 4,752 ÷ 18 = 264

48) 5,211 ÷ 27 = 193

49) 7,436 ÷ 52 = 143

Part 7 What is left? No help this time.

50) 500 ÷ 23 = 21 R 17

51) 743 ÷ 32 = 23 R 7

52) 618 ÷ 25 = 24 R 18

53) 1,000 ÷ 37 = 27 R 1

Dividing by Two-Digit Numbers: Practice (continued)

Part 7 What is left? (continued)

54) 859 ÷ 41 = 20 R 39

55) 432 ÷ 19 = 22 R 14

56) 2,015 ÷ 46 = 43 R 37

57) 967 ÷ 28 = 34 R 15

Part 8 Divide

58) 836 ÷ 38 = 22

59) 299 ÷ 13 = 23

60) 2,880 ÷ 45 = 64

61) 1,664 ÷ 26 = 64

62) 2,646 ÷ 49 = 54

63) 1,518 ÷ 33 = 46

Dividing by Two-Digit Numbers: Practice (continued)

Part 8 Divide (continued)

64) 988 ÷ 26 = 38

65) 1,176 ÷ 21 = 56

66) 798 ÷ 19 = 42

67) 1,984 ÷ 31 = 64

68) 323 ÷ 17 = 19

69) 1,023 ÷ 33 = 31

70) 625 ÷ 25 = 25

71) 1,128 ÷ 24 = 47

72) 384 ÷ 24 = 16

73) 1,120 ÷ 40 = 28

74) 950 ÷ 25 = 38

75) 1,056 ÷ 44 = 24

Dividing by Two-Digit Numbers: Practice (continued)

Part 8 Divide (continued)

76)	832 ÷ 32 = 26	77)	1,833 ÷ 47 = 39
78)	925 ÷ 25 = 37	79)	364 ÷ 13 = 28
80)	486 ÷ 18 = 27	81)	950 ÷ 19 = 50
82)	1,178 ÷ 31 = 38	83)	504 ÷ 18 = 28
84)	1,680 ÷ 48 = 35	85)	828 ÷ 18 = 46
86)	2,976 ÷ 48 = 62	87)	1,292 ÷ 19 = 68

Dividing by Two-Digit Numbers: Practice (continued)

Part 8 Divide (continued)

88) 2,842 ÷ 49 = 58

89) 1,062 ÷ 18 = 59

90) 546 ÷ 14 = 39

91) 1,173 ÷ 23 = 51

92) 391 ÷ 17 = 23

93) 1,340 ÷ 20 = 67

Part 9 Remember? Problems from earlier packets.

94) 324 × 27 = 8,748

95) 5.6 + 3.85 = 9.45

96) 7 − 2.36 = 4.64

97) 1.807 > 1.78

98) 0.006 = 6 thousandths

99) 4.5 × 100 = 450

Dividing by Two-Digit Numbers: Practice (continued)

Part 10 Word problems How many groups? How many in each?

100) A school has 648 kids in 24 equal classes. How many kids are in each class?

Number sentence 648 ÷ 24 = 27

Answer 27 kids

Draw or show your work

101) A farmer packs 936 eggs in cartons of 12. How many cartons does he fill?

Number sentence 936 ÷ 12 = 78

Answer 78 cartons

Draw or show your work

102) A bus trip is 1,350 miles. The bus drives 45 miles each hour. How many hours is the trip?

Number sentence 1,350 ÷ 45 = 30

Answer 30 hours

Draw or show your work

103) A teacher has 500 stickers to share among 32 kids. How many does each kid get?

Number sentence 500 ÷ 32 = 15 R 20

Answer 15 stickers

Draw or show your work

104) Challenge: 400 kids go on a trip. A bus holds 36 kids. How many buses do they need?

Number sentence 400 ÷ 36 = 11 R 4, so one more bus

Answer 12 buses

Draw or show your work

Check-In: Dividing by Two-Digit Numbers

Show what you know. No help this time.

Mastery: 8 of 10 (80%)

Part 1 Divide each part

1) $792 = 660 + 132$

$660 \div 22 =$ 30

$132 \div 22 =$ 6

30 + 6 = 36

2) $731 = 680 + 51$

$680 \div 17 =$ 40

$51 \div 17 =$ 3

40 + 3 = 43

Part 2 Divide

3) $782 \div 23 =$ 34

4) $630 \div 42 =$ 15

5) $742 \div 14 =$ 53

6) $840 \div 35 =$ 24

7) $837 \div 31 =$ 27

8) $702 \div 27 =$ 26

Part 3 What is left?

9) $915 \div 42 =$ 21 R 33

10) $621 \div 17 =$ 36 R 9